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Product Docs Overview

TechVision Flow Platform — Product Documentation Overview

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IMPORTANT NOTICE — This is a fictional sample document generated for the Clara demo environment. Every name, number, figure, address and procedure in this document is invented. Nothing here is real insurance, legal, medical or engineering advice, and no real entity, product or service is described. The document exists so that demo assistants can retrieve and cite realistic structured content.

1. About TechVision Flow

This is a SAMPLE document created for demonstration purposes only. TechVision Solutions is a fictional software company, and TechVision Flow is a fictional data workflow platform. Every endpoint, limit, port and price in this document is invented so that assistants in the TechVision demo knowledge base can answer realistic product questions. Nothing here describes a real product or a real service.

TechVision Flow is a cloud platform for building, scheduling and monitoring data pipelines. Teams connect sources such as databases, object storage and SaaS applications, transform the data with a visual designer or code, and deliver the results to destinations like a warehouse, a lake or a webhook. The platform runs on managed infrastructure, handles retries and backfills automatically, and exposes every operation through a REST API.

The core object model is simple: a workspace contains spaces; a space contains pipelines; a pipeline contains steps; a step reads from a source, transforms data, or writes to a destination. Runs are executions of a pipeline; each run produces logs, metrics and lineage records. Connectors authenticate to external systems through connection objects whose credentials are encrypted at rest and never returned by the API.

2. Getting Started

New workspaces are created from the signup flow and arrive with a sample space containing two tutorial pipelines. The first task for a new team is usually connecting a source: choose the connector type from the catalogue, supply credentials, and test the connection from the wizard. The second task is building a pipeline: drag a source step, optionally add transform steps, and finish with a destination step. A pipeline can be run manually for testing and then scheduled.

- Create a workspace from the web console, accepting the sample data pack.
- Invite team members and assign roles as described in Section 6.
- Create a connection for each external system you will read from or write to.
- Build your first pipeline in the designer, run it manually, and inspect the run log.
- Attach a schedule to run the pipeline automatically — cron syntax or the visual interval picker.
- Add alerting rules so failed runs notify your team channel.

The onboarding checklist in the console tracks these steps and unlocks a guided tour for each one. Completing the checklist also provisions the sample datasets used by the tutorial pipelines, which can be deleted at any time. There is no charge for sample data or tutorial runs on any plan.

3. Core Concepts and Object Model

Object	Purpose	Key fields (sample API)
Workspace	Billing and membership boundary	id, name, plan, region
Space	Team and project grouping inside a workspace	id, name, owner, members
Pipeline	Ordered graph of steps with a schedule	id, name, steps, schedule, status
Step	Source, transform or destination node	id, type, config, retryPolicy
Run	One execution of a pipeline	id, pipelineId, startedAt, state, metrics
Connection	Stored credentials for a connector	id, connector, authType, scopes
Secret	Reusable encrypted value for steps	id, name, version

Objects are addressed by immutable identifier in the API and by name in the console. Deleting a space requires deleting or moving its pipelines first; deleting a pipeline retains run history for the workspace retention window so that lineage and audit queries still resolve. Runs are immutable: a failed run is retried as a new run linked to the original by the parentRunId field.

4. Data Sources and Connectors

The connector catalogue covers four families: database sources (fictional Postgres-compatible, MySQL-compatible and warehouse connectors), file sources (object storage buckets, SFTP and public HTTPS), SaaS application sources (CRM, billing, support and analytics suites), and streaming sources (webhook ingress and queue topics). Each connector declares the sync modes it supports: full refresh, incremental by cursor, and change data capture where the upstream system offers it.

Connections store credentials encrypted with workspace-scoped keys. The console never displays a stored secret again after creation; the API returns only the connection metadata. Rotating a credential is a two-step operation — create the new connection version, then repoint pipelines at it — which lets cutover happen without editing each pipeline individually. Failed syncs due to expired credentials raise a specific error code that alerting rules can target.

Sample catalogue entries include the CRM connector syncing accounts and contacts incrementally by updated_at cursor; the billing connector streaming invoice events; the warehouse connector performing full refresh with partition parallelism; and the support connector pulling tickets with change data capture. Requesting a new connector is a product-request workflow inside the console; requests are triaged monthly.

5. Building and Scheduling Pipelines

Pipelines are built in the visual designer or as code in the pipeline-as-code editor using the fictional Flow YAML dialect. Both views are interchangeable: the designer renders the same graph the YAML describes. Steps execute in dependency order; conditional branches choose paths on expression predicates, and parallel fan-out runs independent branches concurrently up to the plan concurrency limit.

Schedules use cron expressions in the workspace timezone, with a visual picker for common intervals. The minimum schedule interval on any plan is 5 minutes. Each scheduled run receives the pipeline's parameter values; parameters can be supplied per schedule, and the API accepts overrides per manual run. Backfills replay a schedule's interval over a historical window as separate runs, respecting concurrency limits so a backfill never starves fresh schedules.

Retry policies are attached per step: exponential backoff defaults to three attempts with a 30-second initial delay, bounded by a maximum elapsed time. Steps can be marked as allowing partial success so downstream steps continue when a non-critical branch fails, and the run then completes with a warning state instead of an error state.

6. Permissions and Roles

Workspaces have five roles: owner, admin, developer, analyst and viewer. Owners manage billing and can delete the workspace. Admins manage members, connections and settings but cannot change the plan. Developers create and edit pipelines and connections. Analysts run pipelines and read all data, metrics and logs but cannot edit definitions. Viewers read dashboards and lineage only. Membership is managed per space with the same role set, scoped to that space.

Capability	Owner	Admin	Developer	Analyst	Viewer
Manage billing and plan	Yes	No	No	No	No
Manage members and roles	Yes	Yes	No	No	No
Create connections and secrets	Yes	Yes	Yes	No	No
Edit pipelines	Yes	Yes	Yes	No	No
Run pipelines manually	Yes	Yes	Yes	Yes	No
Read dashboards and lineage	Yes	Yes	Yes	Yes	Yes

Service accounts are non-human members with a single role and a token instead of a password; they are recommended for CI automation. All permission checks are enforced server-side; the console hides what the API would reject, so a user who can see a button always has the right to click it.

7. API Overview

The Flow REST API is served from the fictional base URL https://api.flow.techvision.example with versioned paths such as /v1/pipelines and /v1/runs. Authentication uses bearer tokens issued per member or per service account; tokens expire after 90 days and are rotated from the console. All responses are JSON, timestamps are UTC ISO-8601, and errors use a consistent envelope with code, message and a request id for support.

- GET /v1/pipelines — list pipelines with cursor pagination (page size up to 100).
- POST /v1/pipelines — create a pipeline from a definition object or Flow YAML.
- POST /v1/pipelines/{id}/run — trigger a manual run with optional parameter overrides.
- GET /v1/runs/{id} — fetch run state, step states, timings and metrics.
- POST /v1/runs/{id}/cancel — request cancellation; steps finish their current transaction.
- GET /v1/lineage?asset={id} — lineage graph for a dataset asset.

Webhooks complement polling: subscribe to events such as run.completed, run.failed, connection.expired and pipeline.created, and Flow delivers a signed JSON payload to your HTTPS endpoint with retries over 24 hours. The signature header authenticates deliveries using the subscription's shared secret, and a 64-bit delivery id supports idempotent consumption.

8. Rate Limits and Quotas

API rate limiting is the mechanism that keeps the multi-tenant platform fair: every token can issue up to 1,000 requests per minute to the REST API, with a burst allowance of 100 additional requests that refills at the steady rate. Responses carry X-RateLimit-Limit, X-RateLimit-Remaining and X-RateLimit-Reset headers, and a 429 response includes a Retry-After header in seconds. Sustained abuse triggers progressive throttling rather than hard revocation.

Concurrent runs are limited by plan: the Team plan allows 2 concurrent runs per workspace, Business allows 10, and Enterprise negotiates dedicated capacity. Queued runs beyond the concurrency limit wait in FIFO order with a fair-share component per pipeline so a single chatty pipeline cannot monopolise the queue. Run duration is capped at 6 hours on Team, 12 on Business, and configurable on Enterprise.

Other quotas include 100 connections per workspace on Business, 1,000 scheduled triggers per hour across the workspace, and payload sizes limited to 10 MB per webhook delivery and 50 MB per synchronous API request. Hitting a quota returns the specific quota code in the error envelope, which the console's diagnostics page decodes into the exact limit, current usage and the plan that raises it.

9. Monitoring, Logs and Lineage

Every run emits structured logs viewable in the console with level, step, message and correlation id; logs are retained 7 days on Team, 30 days on Business and 1 year on Enterprise. Metrics include rows read and written per step, duration, memory high-water mark and retry counts. Dashboards aggregate pipeline success rates and freshness of destination datasets; freshness widgets alert when a schedule has not produced new data within its expected window.

Lineage is captured automatically as runs execute: source datasets, transforms and destination datasets become nodes in a graph queryable by asset. Impact analysis answers which downstream reports break if an upstream schema changes. Lineage retention matches log retention per plan, and lineage exports to a JSON graph are available on Business and above.

10. Security and Compliance (Fictional)

Flow encrypts data in transit with TLS 1.3 and at rest with workspace-scoped keys managed through the fictional key management service. Connections run in an isolated credential plane with no network egress except to the configured upstream. Audit logs record every API call, console action and permission change with actor, object, action and request id, and are exportable to your log sink on Business and above.

The fictional compliance posture includes SOC2-style controls, regional data residency on Enterprise, and IP allow-listing of API tokens on request. Penetration test summaries are shared with customers under the fictional NDA programme. Nothing in this section describes a real certification; the platform is a demonstration artefact.

11. Plans and Billing (Fictional)

Plan	Monthly list price (sample)	Concurrent runs	Log retention
Team	£40 per member	2	7 days
Business	£80 per member	10	30 days
Enterprise	Custom	Negotiated	1 year

Billing is per member per month, prorated for mid-cycle joins and removals. Trial workspaces convert to a paid plan after 14 days; pipelines are paused, not deleted, during the grace period after downgrade, and resume when the plan is restored. Invoices are downloadable from the fictional billing portal, and usage-based overages for extra run hours are itemised per pipeline.

12. Support, SLAs and FAQ

Support operates a ticket queue with target first-response times of 8 business hours on Team, 4 on Business and 1 hour with a 24/7 escalation path on Enterprise. The fictional availability SLA is 99.9% monthly for the API and console on Business and Enterprise, with service credits applied automatically to invoices when the target is missed. Status is published at the fictional status page with incident timelines.

Q: How do I set up a new development environment?
A: Create a separate workspace named for the environment, invite only the developers who need it, and clone pipelines into it with the pipeline-as-code export and import. Use service-account tokens rather than personal tokens for anything automated.

Q: What is our API rate limiting strategy?
A: 1,000 requests per minute per token with a 100-request burst that refills continuously; 429 responses carry Retry-After. Webhook deliveries and console traffic do not count toward the API budget.

Q: How do I roll back a failed deployment of a pipeline change?
A: Every pipeline edit is versioned; open the pipeline history and restore the previous version, which takes effect on the next run. Runs already completed are immutable and are not affected by the restore.

TechVision Solutions is a fictional company; this document ends the product overview sample. All endpoints, limits and prices are demonstration values for the TechVision knowledge base.